

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-I & II (NEW) EXAMINATION – SUMMER-2019****Subject Code: 3110015****Date: 01/06/2019****Subject Name: Mathematics –2****Time: 10:30 AM TO 01:30 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		Marks
Q.1	(a) Find the Fourier integral representation of $f(x) = \begin{cases} x & ; x \in (0, a) \\ 0 & ; x \in (a, \infty) \end{cases}$	03
	(b) Define: Unit step function. Use it to find the Laplace transform of $f(t) = \begin{cases} (t-1)^2 & ; t \in (0, 1] \\ 1 & ; t \in (1, \infty) \end{cases}$	04
	(c) Use the method of undetermined coefficients to solve the differential equation $y'' - 2y' + y = x^2 e^x$.	07
Q.2	(a) Evaluate $\oint_C \bar{F} \cdot d\bar{r}$; where $\bar{F} = (x^2 - y^2)\hat{i} + 2xy\hat{j}$ and C is the curve given by the parametric equation $C: r(t) = t^2\hat{i} + t\hat{j}; 0 \leq t \leq 2$.	03
	(b) Apply Green's theorem to find the outward flux of a vector field $\bar{F} = \frac{1}{xy}(x\hat{i} + y\hat{j})$ across the curve bounded by $y = \sqrt{x}$, $2y = 1$ and $x = 1$.	04
	(c) Integrate $f(x, y, z) = x - yz^2$ over the curve $C = C_1 + C_2$, where C_1 is the line segment joining (0,0,1) to (1,1,0) and C_2 is the curve $y=x^2$ joining (1,1,0) to (2,4,0).	07
OR		
	(c) Check whether the vector field $\bar{F} = e^{y+2z}\hat{i} + x e^{y+2z}\hat{j} + 2x e^{y+2z}\hat{k}$ is conservative or not. If yes, find the scalar potential function $\phi(x, y, z)$ such that $\bar{F} = \text{grad } \phi$.	07
Q.3	(a) Write a necessary and sufficient condition for the differential equation $M(x, y)dx + N(x, y)dy = 0$ to be exact differential equation. Hence check whether the differential equation $[(x+1)e^x - e^y]dx - xe^y dy = 0$ is exact or not.	03
	(b) Solve the differential equation $(1 + y^2)dx = (e^{-\tan^{-1}y} - x)dy$	04
	(c) By using Laplace transform solve a system of differential equations $\frac{dx}{dt} = 1 - y$, $\frac{dy}{dt} = -x$, where $x(0) = 1, y(0) = 0$.	07
OR		
Q.3	(a) Solve the differential equation $(2x^3 + 4y)dx - xdy = 0$.	03

- (b) Solve: $(x + 1) \frac{dy}{dx} - y = e^{3x}(x + 1)^2$. 04
- (c) By using Laplace transform solve a differential equation $\frac{d^2y}{dt^2} + 5 \frac{dy}{dt} + 6y = e^{-t}$, where $y(0) = 0$, $y'(0) = -1$. 07
- Q.4** (a) Find the general solution of the differential equation 03
 $e^{-y} \frac{dy}{dx} + \frac{e^{-y}}{x} = \frac{1}{x^2}$
- (b) Solve : $\frac{d^3y}{dx^3} - 7 \frac{dy}{dx} + 6y = e^x$ 04
- (c) Find a power series solution of the differential equation $y'' - xy = 0$ near an ordinary point $x=0$. 07
- OR**
- Q.4** (a) Find the general solution of the differential equation 03
 $\frac{dy}{dx} + \frac{y}{x} - \sqrt{y} = 0$.
- (b) Solve : $x^3 \frac{d^3y}{dx^3} + 2x^2 \frac{d^2y}{dx^2} + 2y = x$ 04
- (c) Find a Frobenius series solution of the differential equation $2x^2y'' + xy' - (x + 1)y = 0$ near a regular-singular point $x=0$. 07
- Q.5** (a) Write Legendre's polynomial $P_n(x)$ of degree- n and hence obtain $P_1(x)$ and $P_2(x)$ in powers of x . 03
- (b) Classify ordinary points, singular points, regular-singular points and irregular-singular points (if exist) of the differential equation $y'' + xy' = 0$. 04
- (c) Solve the differential equation 07
 $x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = x^3 \cos x$
 by using the method of variation of parameters.
- OR**
- Q.5** (a) Write Bessel's function $J_p(x)$ of the first kind of order- p and hence show 03
 that $J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x$.
- (b) Classify ordinary points, singular points, regular-singular points and irregular-singular points (if exist) of the differential equation $xy'' + y' = 0$. 04
- (c) Solve the differential equation $y'' + 25y = \sec 5x$ 07
 by using the method of variation of parameters.

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER– I & II (NEW) EXAMINATION – WINTER 2019****Subject Code: 3110015****Date: 01/01/2020****Subject Name: Mathematics –2****Time: 10:30 AM TO 01:30 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

	Marks
Q.1 (a) Find the length of curve of the portion of the circular helix $\vec{r}(t) = \cos t \hat{i} + \sin t \hat{j} + t \hat{k}$ from $t = 0$ to $t = \pi$	03
(b) $\int_{(1,2)}^{(3,4)} (xy^2 + y^3)dx + (x^2y + 3xy^2)dy$ is independent of path joining the points $(1, 2)$ and $(3, 4)$. Hence, evaluate the integral.	04
(c) Verify tangential form of Green's theorem for $\vec{F} = (x - \sin y)\hat{i} + (\cos y)\hat{j}$, where C is the boundary of the region bounded by the lines $y = 0, x = \pi/2$ and $y = x$.	07
Q.2 (a) Find the Laplace transform of $f(t)$ defined as	03
$f(t) = \begin{cases} \frac{t}{k} & 0 < t < k \\ 1 & t > k \end{cases}$	
(b) Find the inverse Laplace transform of $\frac{s^2}{(s^2 + a^2)(s^2 + b^2)}$	04
(c) (i) Calculate the curl of the vector $xyz\hat{i} + 3x^2y\hat{j} + (xz^2 - y^2z)\hat{k}$ (ii) The temperature at any point in space is given by $T = xy + yz + zx$. Determine the derivative of T in the direction of the vector $3\hat{i} - 4\hat{k}$ at the point $(1, 1, 1)$.	07
OR	
(c) Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, $r = \vec{r} $, and $\vec{a}$ is a constant vector. Find the value of $\text{div} \left(\frac{\vec{a} \times \vec{r}}{r^n} \right)$	07
Q.3 (a) Find constants a, b and c such that $\vec{V} = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy + 2z)\hat{k}$ is irrotational.	03
(b) Using Fourier cosine integral representation show that $\int_0^\infty \frac{\cos \omega x}{k^2 + \omega^2} d\omega = \frac{\pi e^{-kx}}{2k}$	04
(c) Solve the following differential equations:	07
(i) $\cos(x + y) dy = dx$	
(ii) $\sec^2 y \frac{dy}{dx} + x \tan y = x^3$	

OR

- Q.3 (a)** Find the Laplace transform of (i) $\int_0^t \frac{\sin t}{t} dt$ (ii) $t^2 u(t-3)$ **03**
- (b)** Using Convolution theorem obtain $L^{-1}\left(\frac{1}{s(s^2 + a^2)}\right)$ **04**
- (c)** Find the power series solution of $\frac{d^2 y}{dx^2} + xy = 0$ **07**
- Q.4 (a)** Find the Laplace transform of the waveform **03**
 $f(t) = \left(\frac{2t}{3}\right), 0 \leq t \leq 3$
- (b)** Using the Laplace transforms, find the solution of the initial value problem **04**
 $y'' + 25y = 10 \cos 5t \quad y(0) = 2, y'(0) = 0$
- (c)** Using variation of parameter method solve $(D^2 + 1)y = x \sin x$ **07**
- OR**
- Q.4 (a)** Solve $y \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = \frac{dy}{dx}$ **03**
- (b)** Solve $y''' - 3y'' + 3y' - y = 4e^t$ **04**
- (c)** Solve $\frac{d^2 y}{dx^2} + 2\frac{dy}{dx} + 4y = 2x^2 + 3e^{-x}$ using method of undetermined coefficients. **07**
- Q.5 (a)** Classify the singular points of the equation $x^3(x-2)y'' + x^3y' + 6y = 0$ **03**
- (b)** Solve $(D^2 + 4)y = \cos 2x$ **04**
- (c)** Solve (i) $ye^x dx + (2y + e^x)dy = 0$ (ii) $\frac{dy}{dx} + 2y \tan x = \sin x$ **07**
- OR**
- Q.5 (a)** Solve $\frac{dy}{dx} = \frac{y^3}{e^{2x} + y^2}$ **03**
- (b)** If $y_1 = x$ is one of solution of $x^2 y'' + xy' - y = 0$ find the second solution. **04**
- (c)** Using Frobenius method solve $x^2 y'' + 4xy' + (x^2 + 2)y = 0$ **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE – SEMESTER 1&2 EXAMINATION – SUMMER 2020****Subject Code: 3110015****Date: 09/11/2020****Subject Name: Mathematics II****Time: 10:30 AM TO 01:30 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		Marks
Q.1	(a) Evaluate $\int_C \vec{F} \cdot d\vec{r}$ along the parabola $y^2 = x$ between the points (0, 0) and (1, 1) where $\vec{F} = x^2\hat{i} + xy\hat{j}$	03
	(b) Find the work done in moving particle from A (1, 0, 1) to B (2, 1, 2) along the straight-line AB in the force field $\vec{F} = x^2\hat{i} + (x - y)\hat{j} + (y + z)\hat{k}$	04
	(c) Verify green's theorem for $\oint_C (2xydx - y^2dy)$ where C is the boundary of the region bounded by the ellipse $3x^2 + 4y^2 = 12$	07
Q.2	(a) Find the Laplace transform of $te^{-4t} \sin 3t$.	03
	(b) Find the inverse Laplace transform of $\frac{5s+3}{(s-1)(s^2+2s+5)}$.	04
	(c) Show that the vector field $\vec{F} = (y \sin z - \sin x)\hat{i} + (x \sin z + 2yz)\hat{j} + (xy \cos z + y^2)\hat{k}$ is conservative and find the corresponding scalar potential.	07
	OR	
	(c) Show that $\vec{F} = 2xyz\hat{i} + (x^2z + 2y)\hat{j} + x^2y\hat{k}$ is irrotational and find a scalar function ϕ such that $\vec{F} = \text{grad}\phi$.	07
Q.3	(a) Find the directional derivative of $f(x, y) = xy + xe^y + \cos(xy)$ at the point $P(1, 0)$ in the direction of $\vec{u} = 3\hat{i} - 4\hat{j}$.	03
	(b) Find the inverse Laplace transform of $\log\left(1 + \frac{1}{s^2}\right)$.	04
	(c) Find the singular solution and general solution of $y + px = x^4 p^2$	07
	OR	
Q.3	(a) Find the Laplace transform of $\frac{\cos at - \cos bt}{t}$.	03
	(b) Show that $\int_0^\infty \frac{\omega^3 \sin \omega x}{\omega^4 + 4} d\omega = \frac{\pi}{2} e^{-x} \cos x; x > 0$.	04
	(c) Find the power series solution of $y' - 2xy = 0; y(0) = 1$ near $x = 0$.	07

- Q.4** (a) Find the Laplace transform of $e^{-t} \{1 - u(t-2)\}$. **03**
- (b) Solve $\frac{d^2x}{dt^2} - 2\frac{dx}{dt} + x = e^t$ with $x = 2, \frac{dx}{dt} = -1$ at $t = 0$. **04**
- (c) Solve $(D^2 - 1)y = xe^x \sin x$ **07**
- OR**
- Q.4** (a) Solve $\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$ **03**
- (b) Using method of variation of parameter, solve $\frac{d^2y}{dx^2} + 4y = \tan 2x$. **04**
- (c) Using method of undetermined coefficients solve **07**
 $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = x^2 e^x$.
- Q.5** (a) Classify the singular points of $x^2 y'' + xy' - 2y = 0$. **03**
- (b) Solve $\frac{d^2y}{dx^2} + 9y = \sin 2x \sin x$. **04**
- (c) Solve (i) $(x^3 + 3xy^2)dx + (3x^2y + y^3)dy = 0$. **07**
(ii) $\frac{dy}{dx} + y \cot x = 2 \cos x$.
- OR**
- Q.5** (a) Solve $\frac{dy}{dx} = \frac{y + \sqrt{x^2 + y^2}}{x}$. **03**
- (b) Solve $x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + y = \cos(\ln x)$. **04**
- (c) Using Frobenius method solve $2x^2 y'' + xy' - (x+1)y = 0$. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-1/2 EXAMINATION – WINTER 2021****Subject Code:3110015****Date:21/03/2022****Subject Name:Mathematics - 2****Time:10:30 AM TO 01:30 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Find $L\{t^3 e^{-4t}\}$.	03
	(b) Find $L^{-1}\left\{\frac{6e^{-2s}}{s^2 + 4}\right\}$.	04
	(c) Verify Green's theorem for the function $\bar{F} = (x + y)i + 2xyj$ and C is the rectangle in the xy -plane bounded by $x = 0, y = 0, x = a, y = b$.	07
Q.2	(a) Find $L\{te^{4t} \cos 2t\}$.	03
	(b) Find the Fourier cosine integral of $f(x) = \frac{\pi}{2} e^{-x}, x \geq 0$.	04
	(c) (i) Find the directional derivative of $f(x, y, z) = 2x^2 + 3y^2 + z^2$ at the point $(2, 1, 3)$ in the direction of $\bar{a} = (1, 0, -2)$.	03
	(ii) If $\bar{F} = (2y + 3)i + xzj + (yz - x)k$, evaluate $\int_C \bar{F} \cdot d\bar{r}$ along the path C: $x = 2t^2, y = t, z = t^3$ from $t=0$ to $t=1$.	04
	OR	
	(c) Solve in series $3xy'' + 2y' + y = 0$ using Frobenius method.	07
Q.3	(a) Find the arc length of the curve (semi-circular) $x(t) = \cos t, y(t) = \sin t, z(t) = 0; 0 \leq t \leq \pi$.	03
	(b) A vector field is given by $\bar{F} = (x^2 + xy^2)i + (y^2 + x^2y)j$. Show that $\bar{F}$ is irrotational and find its scalar potential.	04
	(c) Use divergence theorem for $\bar{F} = (x^2 - yz)i + (y^2 - zx)j + (z^2 - xy)k$ over the surface of rectangular parallelepiped, $0 \leq x \leq a, 0 \leq y \leq b, 0 \leq z \leq c$ to evaluate $\iiint_S \bar{F} \cdot \hat{n} ds$.	07
	OR	
Q.3	(a) Solve $\frac{dy}{dx} - y \cot x = 2x \sin x$.	03
	(b) Solve $y'' + y' - 12y = e^{6x}$.	04
	(c) Solve $\frac{dy}{dt} - 4y = 2e^{2t} + e^{4t}$ by Laplace transformation.	07

- Q.4** (a) Solve $\frac{dy}{dx} + \frac{y}{x} = y^3$. **03**
- (b) Solve $(x^2 - 4xy - 2y^2)dx + (y^2 - 4xy - 2x^2)dy = 0$. **04**
- (c) Solve $y'' + 9y' = 2x^2$ using the method of undetermined coefficients. **07**
- OR**
- Q.4** (a) Solve $4xp^2 = (3x - a)^2$. **03**
- (b) Solve $x^2 y'' + xy' - 4y = x^2$. **04**
- (c) (i) Express $2 - 3x + 4x^2$ in terms of Legendre's polynomial. **03**
- (ii) Find ordinary and singular points of $2x^2 y'' + 6xy' + (x + 3)y = 0$. **04**
- Q.5** (a) Solve $(y - px)(p - 1) = p$. **03**
- (b) Solve $(D^3 + D)y = \cos x$. **04**
- (c) Solve $y'' + 4y = \sec 2x$ by using the method of variation of parameters. **07**
- OR**
- Q.5** (a) Solve $(D^3 - 6D^2 + 11D - 6)y = 0$. **03**
- (b) Solve $(2x + 3)^2 y'' - 2(2x + 3)y' - 12y = 6x$. **04**
- (c) Find the series solution of $(1 + x^2)y'' + xy' - 9y = 0$ near the ordinary point $x=0$. **07**

Seat No.: _____

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GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I & II(NEW)EXAMINATION – SUMMER 2022

Subject Code:3110015

Date:22-08-2022

Subject Name:Mathematics - 2

Time:10:30 AM TO 01:30 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	Marks
Q.1 (a) Find the Laplace transform of $t^2 e^{-3t}$.	03
(b) Define conservative vector field and potential function.	04
(c) Solve $y''' - 3y'' + 3y' - y = 4e^x$ using the method of undetermined coefficients.	07
Q.2 (a) Find the divergence of $\mathbf{F} = (x^2 - y)\mathbf{i} + (xy - y^2)\mathbf{j}$.	03
(b) Find Fourier cosine integral of $f(x) = e^{-kx} (x > 0, k > 0)$	04
(c) Integrate $f(x, y, z) = 3x^2 - 2y + z$ over the line segment C joining the origin to the point $(2, 2, 2)$.	07
OR	
(c) Write Green's theorem. Evaluate the integral $\oint_C \{xydy - y^2dx\}$ where C is the square cut from the first quadrant by the lines $x = 1$ and $y = 1$.	07
Q.3 (a) Obtain convolution of t and e^t .	03
(b) Find the Laplace transform of $\frac{\cos at - \cos bt}{t}$.	04
(c) Solve the initial value problem $y'' - y' - 2y = 0, y(0) = 1, y'(0) = 0$ using Laplace transform.	07
OR	
Q.3 (a) Find the inverse Laplace transform of $\frac{s-4}{s^2-4}$.	03
(b) State second shifting theorem and find the inverse Laplace transform of the function $\frac{se^{-\pi s}}{s^2+1}$.	04
(c) State convolution theorem and using it obtain the inverse Laplace transform of $\frac{1}{s(s^2+4)}$.	07
Q.4 (a) Solve $\frac{dy}{dx} - 2y = 4 - x$.	03
(b) Solve $p^2 + 2p \cot x = y^2$.	04
(c) Solve $y'' + 4y = 4 \tan 2x$ using the method of variation of parameters.	07
OR	
Q.4 (a) Find particular solution of $y'' - 2y' + y = \cos 3x$.	03
(b) Solve $x^2 y'' - 3xy' + 4y = 0$	04

- (c) Solve the initial value problem **07**
 $y''' + y' = 0,$
 $y(0) = 0, y'(0) = 1, y''(0) = 2$
- Q.5** (a) Write Legendre's and Bessel's differential equations. **03**
 (b) Solve the differential equation **04**
 $(y \cos x + 2xe^y) + (\sin x + x^2 e^y - 1)y' = 0$
 (c) Find the power series solution of the equation $(x^2 + 1)y'' + xy' -$ **07**
 $xy = 0$ in powers of x .
- OR**
- Q.5** (a) Write Legendre polynomials of degree one and two. **03**
 (b) Solve $y = 2px + p^2y$. **04**
 (c) Solve $x(x - 1)y'' + (3x - 1)y' + y = 0$ about $x = 0$ using Frobenius **07**
 method.
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Seat No.: _____

Enrolment No. _____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER-I & II(NEW) EXAMINATION – WINTER 2022

Subject Code:3110015

Date:14-03-2023

Subject Name:Mathematics - 2

Time:10:30 AM TO 01:30 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- Q-1**
- (a) If $\vec{F} = x^3\hat{i} + y^3\hat{j} + z^3\hat{k}$, find $\text{curl}\vec{F}$. 3
- (b) Solve $x\frac{dy}{dx} + y + 1 = 0$ 4
- (c) Evaluate $\int_C [(2xy - x^2)dx + (x + y^2)dy]$ using Green's theorem where C is closed curve formed by $y = x^2$ and $x = y^2$. 7
- Q-2**
- (a) Find the Laplace transform of $t^2 - e^{-2t} + \cosh^2 3t$. 3
- (b) Find the Fourier cosine integral of $f(x) = \frac{\pi}{2}e^{-x}, x \geq 0$. 4
- (c) Using convolution theorem. Find the inverse Laplace transform of $\frac{1}{(s^2+4)^2}$. 7
- OR**
- (c) If $\vec{F} = 2xyz\hat{i} + (x^2z + 2y)\hat{j} + x^2y\hat{k}$ then 7
- (i) If $\vec{F}$ is conservative, find its scalar potential ϕ .
- (ii) Find the work done in moving a particle under this force field from (0,1,1) to (1,2,0).
- Q-3**
- (a) Find the constant b such that 3
- $\vec{A} = (bx + 4y^2z)\hat{i} + (x^3\sin z - 3y)\hat{j} - (e^x + 4\cos x^2y)\hat{k}$ is solenoidal.
- (b) Evaluate $\int_0^\infty te^{-2t}\cos t dt$. 4
- (c) By Laplace transform solve, $\frac{dy}{dt} + y = \cos 2t, y(0) = 1$. 7
- OR**
- Q-3**
- (a) State Second Shifting theorem. Hence find the inverse Laplace transform of $\frac{e^{-as}}{s}$. 3
- (b) Solve $\frac{dy}{dx} + \frac{1}{x} = \frac{e^y}{x^2}$. 4
- (c) (i) Solve $y = 2px - xp^2$ 7
- (ii) Solve $p = \log(px - y)$

- Q-4** (a) If $y_1 = e^{-x}$ is one of the solutions of $y'' + 2y' + y = 0$, then find the second solution. 3
- (b) Solve $(D^2 - 2D + 1)y = 10e^x$. 4
- (c) Find the solution of differential equation $y'' + 4y = 2\sin 3x$ by the method of Undetermined Coefficients. 7

OR

- Q-4** (a) Solve $2D^2y + Dy - 6y = 0$. 3
- (b) Solve $(x^2D^2 - 7xD + 12)y = x^2$. 4
- (c) Using the method of variation of parameter, solve $\frac{d^2y}{dx^2} + y = \sin x$. 7

- Q-5** (a) Find the ordinary points of $(x^2 + 1)\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} + 6y = 0$ 3
- (b) Solve $(D^2 + 1)y = \sin^2 x$. 4
- (c) Find power series solution of $\frac{d^2y}{dx^2} + xy = 0$. 7

OR

- Q-5** (a) Find the value of $J_{\frac{3}{2}}(x)$. 3
- (b) Find the Laplace transform of $f(t) = \frac{2}{3}t, 0 \leq t \leq 3$, if $f(t) = f(t + 3)$. 4
- (c) Solve in series the differential equation $4x\frac{d^2y}{dx^2} + 2y\frac{dy}{dx} + y = 0$. 7

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II(NEW) EXAMINATION – SUMMER 2023****Subject Code:3110015****Date:07-08-2023****Subject Name:Mathematics - 2****Time:10:30 AM TO 01:30 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Find the directional derivative of $f(x, y, z) = x^2 + 5y^2 + 3z^2$ at point (1,1,1) in the direction $3i+4j+5k$.	03
	(b) A vector field is given by $F = 3x^2yz \, i + x^3z \, j + x^3y \, k$, show that f is irrotational and find the scalar ϕ such that $F = \nabla\phi$.	04
	(c) Verify the Greens theorem for $F = x^2i + xy^2j$, along the square bounded by $x = 0, \quad x = 1, \quad y = 0 \text{ and } y = 1$	07
Q.2	(a) Discuss about ordinary point, singular point, regular singular point and irregular singular point of differential equation $(x^2 - 1)y'' +$ $3xy + 5x^2y = 0$	03
	(b) Express $f(x) = x^3 + 2x^2 + 3x - 1$ in terms of Legendre's polynomial.	04
	(c) Find a power series solution of $y'' - 25y = 0$ near an ordinary point $x=0$.	07
	OR	
	(c) Prove that $P_n(x) = \frac{1}{n!2^n} \frac{d^n}{dx^n} (x^2 - 1)^n$	07
Q.3	(a) Find the Laplace transform of (i) $e^{3t}t^5$ (ii) $t \cos 3t$ (iii) $(\sin^2 5t)/t$	03
	(b) Find the Inverse Laplace transform of (i) $\frac{1}{(s-1)(s-2)(s-3)}$ (ii) $\frac{1}{s^4-9s^2}$ (iii) $\frac{s+1}{s^2+2s+10}$ (iv) $\tan^{-1}(\frac{s}{4})$	04
	(c) Solve the initial value problem using Laplace transformation $y'' -$ $5y' - 6y = e^{3t}$ with $y(0) = 3 \text{ and } y'(0) = 2$	07
	OR	
Q.3	(a) Find the Laplace transform of (i) $e^{4t}u(t-2)$ (ii) $(t^2 + 1)u(t-1)$ (iii) $\sin 3t u(t-\pi)$	03
	(b) Using convolution theorem find the Inverse Laplace transform of $\frac{1}{(s^2-4)(s^2-9)}$	04
	(c) Find the Fourier integral of	07

$$f(x) = \begin{cases} 0, & |x| < 5 \\ 5, & |x| > 5 \end{cases}$$

- Q.4 (a)** Solve the Differential equation **03**
 $e^x \cos y \, dx - e^x \sin y \, dy = 0$
- (b)** Solve $\frac{d^2 y}{dx^2} + 6 \frac{dy}{dx} + 9y = 0$, $y(0) = 1$, $y'(0) = 2$ **04**
- (c)** Solve **07**
 (i) $\frac{dy}{dx} + y \tan x = \sin 2x$, $y(0) = 2$
 (ii) $(x^3 + y^3)dx - xy^2 dy = 0$

OR

- Q.4 (a)** Solve $(x^2 - y^2)dx + xydy = 0$ **03**
(b) Solve $\frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} - 14y = 0$, $y(0) = 3$, $y'(0) = 1$ **04**
- (c)** Solve **07**
 (i) $(y - px)(p^2 + 1) = \tan^{-1} p$
 (ii) $p^2 x^2 = x^2 + p^2$
- Q.5 (a)** Solve $\frac{d^2 y}{dx^2} + \frac{dy}{dx} - 12y = e^{-4x} - 12x$ **03**
(b) Using method of undetermined coefficient obtain the solution of **04**
 $\frac{d^2 y}{dx^2} + 4y = \sin x$
- (c)** Find the solution of differential equations **07**
 (i) $\frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = x^2 e^x$
 (ii) $\frac{x^2 d^2 y}{dx^2} - 3x \frac{dy}{dx} + 4y = \cos(\log x)$

OR

- Q.5 (a)** Solve **03**
 $\frac{d^4 y}{dx^4} + 8 \frac{d^2 y}{dx^2} + 16y = \cos 2x$
- (b)** Using method of undetermined coefficient **04**
 $\frac{d^2 y}{dx^2} + \frac{dy}{dx} - 6y = x^2$
- (c)** (i) Using method of variation of parameters find the solution of **07**
 differential equation $\frac{d^2 y}{dx^2} + 16y = \cot 4x$
 (ii) Find the solution of differential equation
 $\frac{x^2 d^2 y}{dx^2} - 6x \frac{dy}{dx} + 6y = x^2 + \frac{1}{x^2}$

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-I (NEW) EXAMINATION – WINTER 2023****Subject Code:3110015****Date:23-01-2024****Subject Name: Mathematics - 2****Time: 02:30 PM TO 05:30 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- | | MARKS |
|---|-----------|
| Q.1 (a) Define Solenoidal Vector field. Find the constant a such that the vector $(x + 3)\hat{i} + (y - 2z)\hat{j} + (x + az)\hat{k}$ is solenoidal. | 03 |
| (b) If $\vec{F} = 3x^2y\hat{i} + (x^3 - 2yz^2)\hat{j} + (3z^2 - 2y^2z)\hat{k}$, is conservative, then find | 04 |
| (i) its scalar potential ϕ . | |
| (ii) the work done in moving from (2,1,1) to (3,0,1). | |
| (c) Verify Green's theorem in the plane for $\oint_C (xy^2 - 2xy)dx + (x^2y + 3)dy$, where C is the region bounded by the rectangle with vertices $(-1,0)$, $(1,0)$, $(1,1)$ and $(-1,1)$. | 07 |
| Q.2 (a) State first shifting theorem for Laplace transform. | 03 |
| Evaluate (i) $L[e^{-t}t^5]$ (ii) $L^{-1}\left[\frac{1}{s^2+2s+2}\right]$. | |
| (b) Find (i) $L\left[\frac{1-\cos 2t}{t}\right]$ (ii) $L^{-1}\left[\frac{54}{(s^2+9)(s^2-9)}\right]$. | 04 |
| (c) Using Laplace transform solve the initial value problem: | 07 |
| $y'' - y = t$, with $y(0) = 1, y'(0) = -1$. | |
| OR | |
| (c) Find the Fourier sine integral of $f(x) = e^{-bx}$ and show that | 07 |
| $\frac{\pi}{2} e^{-bx} = \int_0^\infty \frac{\lambda \sin \lambda x}{\lambda^2 + b^2} d\lambda.$ | |
| Q.3 (a) Define linear differential equation. Solve: $\frac{dy}{dx} + y = e^{-x}$, given that $y(1) = 1$. | 03 |
| (b) Solve: (i) $(D^2 + 7D - 18)y = 0$, (ii) $y'' + y' + 2y = 0$. | 04 |
| (c) (i) Solve: $(x + y - 1)dx + (2x + 2y - 3)dy = 0$,
(ii) Solve: $x^2 p^2 - 4y^2 = 0$. | 07 |
| OR | |
| Q.3 (a) Solve: $y^2 \frac{dx}{dy} + yx = y^3 x^2$. | 03 |
| (b) Determine a second solution of $x^2 \frac{d^2y}{dx^2} - 2y = 0, x > 0$, where first solution is $y_1 = \frac{1}{x}$. | 04 |

- (c) (i) Solve: $p = \sin(y - px)$, 07
(ii) Solve: $(x^3 + y^3)dx - xy^2dy = 0$.

- Q.4** (a) Solve: $(D^2 + 3D + 2)y = \sin 2x$. 03
(b) Solve: $x^3y''' + x^2y'' = x^3$. 04
(c) Using the method of undetermined coefficients solve $y'' + 9y = 2x^2$. 07

OR

- Q.4** (a) Solve: $(D^2 + 4D)y = x + x^2$. 03
(b) Solve: $(D^2 - 6D + 9)y = x^2e^{3x}$. 04
(c) Using the method of variation of parameters solve: $y'' - 2y' + y = xe^x \sin x$. 07

- Q.5** (a) Determine the singular points of the differential equation $2x(x - 2)^2 y'' + 3xy' + (x - 2)y = 0$ and classify them as regular or irregular. 03
(b) Find the series solution of $y'' - y' = 0$. 04
(c) Solve the differential equation $3xy'' - (x - 2)y' + 2y = 0$ by using Frobenius method. 07

OR

- Q.5** (a) Discuss about the ordinary points, regular singular points and irregular singular points for the differential equation $x^3y'' + 5xy' + 3y = 0$. 03
(b) Find the power series solution of $y' + 2xy = 0$ in powers of x . 04
(c) (i) Express $J_3(x)$ in terms of $J_1(x)$ and $J_2(x)$. 07
(ii) Express $f(x) = 4x^3 + 6x^2 + 7x + 1$ in terms of Legendre's polynomials.

Enrolment No./Seat No _____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER-I & II (NEW) EXAMINATION – SUMMER 2024

Subject Code:3110015

Date:02-07-2024

Subject Name:Mathematics - 2

Time:02:30 PM TO 05:30 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

Q.1 (a) Find curl of $\vec{v} = (xyz)\hat{i} + (3x^2y)\hat{j} + (xz^2 - y^2z)\hat{k}$ at $(2, -1, 1)$. **03**

(b) If a force $\vec{F} = 2x^2y\hat{i} + 3xy\hat{j}$ displaces a particle in the xy – plane from $(0,0)$ to $(1,4)$ along a curve $y = 4x^2$. Find the work done. **04**

(c) State and apply Green's theorem to evaluate $\int_C [(2x^2 - y^2)dx + (x^2 + y^2)dy]$, **07**
where C is the boundary of the area enclosed by the x – axis and the upper half of the circle $x^2 + y^2 = a^2$.

Q.2 (a) Find Laplace transform of $f(t) = \int_0^t \frac{\sin t}{t} dt$. **03**

(b) Find the Fourier cosine integral of $f(x) = e^{-kx}$, where $x > 0, k > 0$ **04**

(c) State convolution theorem and use it to find inverse Laplace transform of **07**
 $\frac{1}{(s^2 + a^2)^2}$.

OR

(c) Using Laplace transform solve the following initial value problem **07**
 $y'' + 4y' + 8y = 1, y(0) = 0, y'(0) = 1$.

Q.3 (a) Solve $\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 9x = 0$. **03**

(b) Find the inverse Laplace transform of $\frac{se^{-\frac{s}{2}} + \pi e^{-s}}{s^2 + \pi^2}$. **04**

(c) Solve (i) $y + px = x^4 p^2$ **07**
(ii) $p^2 - xp + y = 0$

OR

Q.3 (a) Solve $(y^2 - x^2)dx + 2xy dy = 0$ **03**

(b) Find the Laplace transform of the waveform **04**
 $f(t) = \left(\frac{2t}{3}\right), 0 \leq t \leq 3$.

- (c) Find the series solution of $(1+x^2)y'' + xy' - 9y = 0$. 07
- Q.4** (a) Solve $9yy' + 4x = 0$. 03
- (b) If $y_1 = x$ is one of the solution of $x^2y'' + xy' - y = 0$, find the second solution. 04
- (c) Using the method of variation of parameter, solve $\frac{d^2y}{dx^2} + y = \sin x$. 07
- OR
- Q.4** (a) Find Laplace transform of $t^2u(t-2)$. 03
- (b) Solve $(D^2 + 9)y = 2\sin 3x + \cos 3x$. where $D = \frac{d}{dx}$ 04
- (c) Using the method of undetermined coefficients, solve $y'' - 2y' + 5y = 5x^3 - 6x^2 + 6x$. 07
- Q.5** (a) Solve $x^2y'' - 20y = 0$. 03
- (b) Solve $(D^2 - 1)y = xe^x$ where $D = \frac{d}{dx}$ 04
- (c) Using Frobenius method, solve $4x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$. 07
- OR
- Q.5** (a) Classify the singular points of the equation $x^3(x-2)y'' + x^3y' + 6y = 0$. 03
- (b) Prove that $\frac{d}{dx}[J_n^2(x)] = \frac{x}{2n}[J_{n-1}^2(x) - J_{n+1}^2(x)]$. 04
- (c) Show that $\int_{-1}^1 x^2 P_{n-1}(x) P_{n+1}(x) dx = \frac{2n(n+1)}{(2n-1)(2n+1)(2n+3)}$. 07

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II EXAMINATION – WINTER 2024****Subject Code:3110015****Date:10-01-2025****Subject Name:Mathematics - 2****Time:10:30 AM TO 01:30 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- Q.1** (a) Find a such that $(x+3y)\hat{i} + (y-2z)\hat{j} + (x+az)\hat{k}$ is solenoidal. **03**
- (b) Solve $ye^x dx + (2y + e^x)dy = 0$ **04**
- (c) Verify Green's theorem in a plane for the integral $\int_C [(x-2y)dx + xdy]$ taken **07**
around the circle $x^2 + y^2 = 4$.

- Q.2** (a) Find the Laplace transform of $(\sin 2t - \cos 2t)^2$. **03**
- (b) Find the Fourier sine integral of $f(x) = e^{-bx}$. **04**
- (c) Using the Frobenius method, obtain the series solution for **07**
 $2x(1-x)y'' + (1-x)y' + 3y = 0$ about $x_0 = 0$.

OR

- (c) Using the method of undetermined coefficients, solve **07**
 $(D^2 - 9)y = x + e^{2x} - \sin 2x$.
- Q.3** (a) Find the arc length of the curve $\vec{r}(t) = t^2\hat{i} + t^3\hat{j}$ between (1,1) and (4,8). **03**
- (b) Find the Laplace transform of $\frac{e^{-t} \sin t}{t}$. **04**
- (c) State the convolution theorem and verify it for $f(t) = t$ and $g(t) = e^{2t}$. **07**

OR

- Q.3** (a) Find the inverse Laplace transform of $\tan^{-1} s$. **03**
- (b) Solve $x^2 p^2 + 3xyp + 2y^2 = 0$. **04**
- (c) Solve the initial value problem using Laplace transformation $y'' - 3y' + 2y = 4t$ **07**
with $y(0) = 1, y'(0) = -1$.

- Q.4** (a) Find the inverse Laplace transform of $\frac{e^{-\pi s}}{s^2 - 2s + 2}$. **03**
- (b) Solve $(D^2 + 1)y = e^{-x}$. **04**
- (c) Using method of variation of parameters, solve $(D^2 - 2D + 2)y = e^x \tan x$. **07**

OR

- Q.4** (a) Classify the singular points of the equation $x^3(x-2)y'' + x^3y' + 6y = 0$. **03**

- (b) Find the Laplace transform of $\sin \sqrt{t}$. 04
- (c) Find the series solution of $(1+x^2)y'' + xy' - 9y = 0$. 07
- Q.5** (a) Solve $dr + (2r \cot \theta + \sin 2\theta)d\theta = 0$ 03
- (b) If $y_1 = \frac{\sin x}{x}$ is one of the solution of $xy'' + 2y' + xy = 0$, find the second solution. 04
- (c) Show that (i) $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$ (ii) $J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \cos x$. 07
- OR**
- Q.5** (a) Find the Laplace transform of $\int_0^t \int_0^t \sin at \, dt \, dt$. 03
- (b) Solve $(x^2 D^2 - xD + 2)y = 6$. 04
- (c) Prove that $J_0^2(x) + 2[J_1^2(x) + J_2^2(x) + J_3^2(x) + \dots] = 1$ 07

Enrolment No./Seat No _____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I&II EXAMINATION – SUMMER 2025

Subject Code:3110015

Date:07-07-2025

Subject Name:Mathematics - 2

Time:10:30 AM TO 01:30 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

Q.1 (a) Show that 03

$\vec{F} = (y^2 - z^2 + 3yz - 2x)\vec{i} + (3xz + 2xy)\vec{j} + (3xy - 2xz + 2z)\vec{k}$ is both solenoidal and irrotational.

(b) Find Fourier integral representation of function $f(x) = \begin{cases} 1 & |x| < 1 \\ 0 & |x| > 1 \end{cases}$ 04

Hence, Evaluate $\int \frac{\sin \omega \cos \omega x}{\omega} d\omega$.

(c) Verify Green's theorem in the plane for $\int_C [(3x^2 - 8y^2)dx + (4y - 6xy)dy]$, where C is the boundary of the region bounded by $x = 0, y = 0, x + y = 1$. 07

Q.2 (a) Solve $ye^x dx + (2y + e^x)dy = 0$ 03

(b) Find the directional Derivative of $\phi = xy^2 + yz^2$ at the point $(2, -1, 1)$ in the direction of the vector $\hat{i} + 2\hat{j} + 2\hat{k}$. 04

(c) Find the power series solution of $\frac{d^2y}{dx^2} + xy = 0$. 07

OR

(c) Find the series solution of $(1 - x^2)y'' - 2xy' + 2y = 0$ 07

Q.3 (a) Solve $(4D^2 - 4D + 1)y = e^{\frac{x}{2}}$ 03

(b) Solve $(D^2 + 6D + 8)y = \cos^2 x$ 04

(c) Solve $(D^2 - 2D + 3)y = x^3 + \sin x$ by using the method of undetermined coefficients 07

OR

Q.3 (a) Solve $(D^2 + a^2)y = \cos ax$ 03

(b) Solve $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = e^x \sin x + 7$ 04

(c) Solve $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = \log x$. $\sin(\log x)$ by Cauchy's linear equation 07

- Q.4 (a)** Find the Laplace Transform Of $f(t) = \sinh at$. **03**
- (b)** Find the Laplace Transform of $te^{4t} \cos 2t$. **04**
- (c)** Solve the differential equation using initial value problem $y'' + 6y' = 1, y(0) = 2, y'(0) = 0$. **07**
- OR**
- Q.4 (a)** Find the inverse Laplace transform of $\log \frac{(s^2+b^2)}{(s^2+a^2)}$ **03**
- (b)** Find the Laplace Transform of $\frac{e^{-2t} \sin 2t \cosh t}{t}$ **04**
- (c)** Find the Inverse Laplace Transform Of $\frac{5s+3}{(s-1)(s^2+2s+5)}$ **07**
-
- Q.5 (a)** Solve $\frac{dy}{dx} + y \cot x = 2 \cos x$ **03**
- (b)** Solve $3x^4 p^2 - xp - y = 0$ **04**
- (c)** Using the method variation of parameter solve the differential equation $(D^2 + 1)y = x \sin x$ **07**
- OR**
- Q.5 (a)** Solve $(xy - 2y^2)dx - (x^2 - 3xy)dy = 0$ **03**
- (b)** Solve $\frac{dy}{dx} + \frac{1}{x}y = \frac{y^2}{x^2}$ **04**
- (c)** Solve $(D^4 - 16)y = e^{2x} + x^4$ **07**

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GUJARAT TECHNOLOGICAL UNIVERSITY

Bachelor of Engineering - SEMESTER - 1/2 EXAMINATION - WINTER 2025

Subject Code: 3110015

Date: 03-01-2026

Subject Name: Mathematics - 2

Time: 02:30 PM TO 05:30 PM

Total Marks: 70

Instructions

2. Make suitable assumptions wherever necessary.

3. Figures to the right indicate full marks.

	Marks
Q.1 (a) If ϕ is a scalar field, prove that $\text{curl}(\text{grad } \phi) = 0$.	03
(b) Determine whether the vector field $\vec{u} = y^2\hat{i} + 2xy\hat{j} - z^2\hat{k}$ is solenoidal or irrotational at the point (1,2,1)?	04
(c) Verify Green's theorem in the plane for $\vec{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$ taken around the rectangle bounded by the lines $x = \pm 1, y = 0$ and $y = 2$.	07
Q.2 (a) Define order and degree of the ordinary differential equation. Give an example of ordinary differential equation: (i) first order first degree, (ii) first order higher degree.	03
(b) Find the Laplace transform of (i) $t \cos^2 t$, (ii) $\frac{\sin t}{t}$.	04
(c) Apply the method of undetermined coefficients to solve the differential equation $(D^2 - 2D + 5)y = 5x^3 - 6x^2 + 6x$.	07
OR	
(c) Apply the variation of parameters method to solve the differential equation $(D^2 - 3D + 2)y = e^x$.	07
Q.3 (a) State first shifting theorem for Laplace transform. Find the Laplace transform of $e^{-3t}(t^2 + \sin t)$.	03
(b) Find the general solution of the differential equation $x^2y dx - (x^3 + xy^2)dy = 0$.	04
(c) Apply Laplace transform to solve the initial value problem: $y'' - 6y' + 9y = t^2 e^{3t}, y(0) = 2, y'(0) = 6$.	07
OR	
(a) State convolution theorem for Laplace transform. Find the inverse Laplace transform of $\frac{1}{s(s+1)}$.	03
(b) Check whether the given differential equation is exact or not: $(x^4 - 2xy^2 + y^4)dx - (2x^2y - 4xy^3 + \sin y) dy = 0$.	04
(c) Find the Fourier cosine integral representation of $f(x) = \frac{\pi}{2}e^{-x}, x > 0$.	07

- Q.4** (a) Find the general solution of the differential equation: $y' + 2y \tan x = \sin x$. 03
- (b) Solve the Euler-Cauchy differential equation : $(4x^2D^2 + 16xD + 9)y = 0$. 04
- (c) Solve: (i) $(D^2 - 4)y = e^{2x} + e^{-4x}$. 07
(ii) $(D^3 - 3D^2 - D + 3)y = 1$.

OR

- (a) Solve: $x^2p^2 + xyp - 6y^2 = 0$. 03
- (b) If $y_1 = x$ is one solution of $x^2y'' + xy' - y = 0$, find the second solution. 04
- (c) Solve (i) $(D^2 + 9)y = \cos 4x$, (ii) $(D^3 - 5D^2 + 8D - 4)y = e^{-x}$. 07
- Q.5** (a) Classify the singular points of the differential equation $x^2y'' + xy' - 2y = 0$. 03
- (b) Find the series solution of $y'' = 2y'$ in powers of x . 04
- (c) Using the power-series method, solve $(1 - x^2)y'' - 2xy' + 2y = 0$. 07

OR

- (a) Discuss about ordinary point, singular point, regular singular point and irregular singular point for the differential equation $x^3(x - 1)y'' + 3(x - 1)y' + 7xy = 0$. 03
- (b) Derive recurrence relation for the Bessel's polynomial $I_n(x)$. 04
- (c) Find the series solution of $2x(x - 1)y'' - (x + 1)y' + y = 0$ at $x = 0$. 07
